```
from google.colab import files
uploaded = files.upload()
```

```
Choose Files dog.jpg
dog.jpg(image/jpeg) - 3772 bytes, last modified: 10/3/2026 - 100% done
Saving dog.jpg to dog.jpg
```

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
filename = list(uploaded.keys())[0]
image = cv2.imread(filename)
print("Image uploaded successfully:", filename)
```

```
Image uploaded successfully: dog.jpg
```

```
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
```

```
sobel_x = cv2.Sobel(
 gray,
 cv2.CV_64F,
 1,
 0,
 ksize=3
)

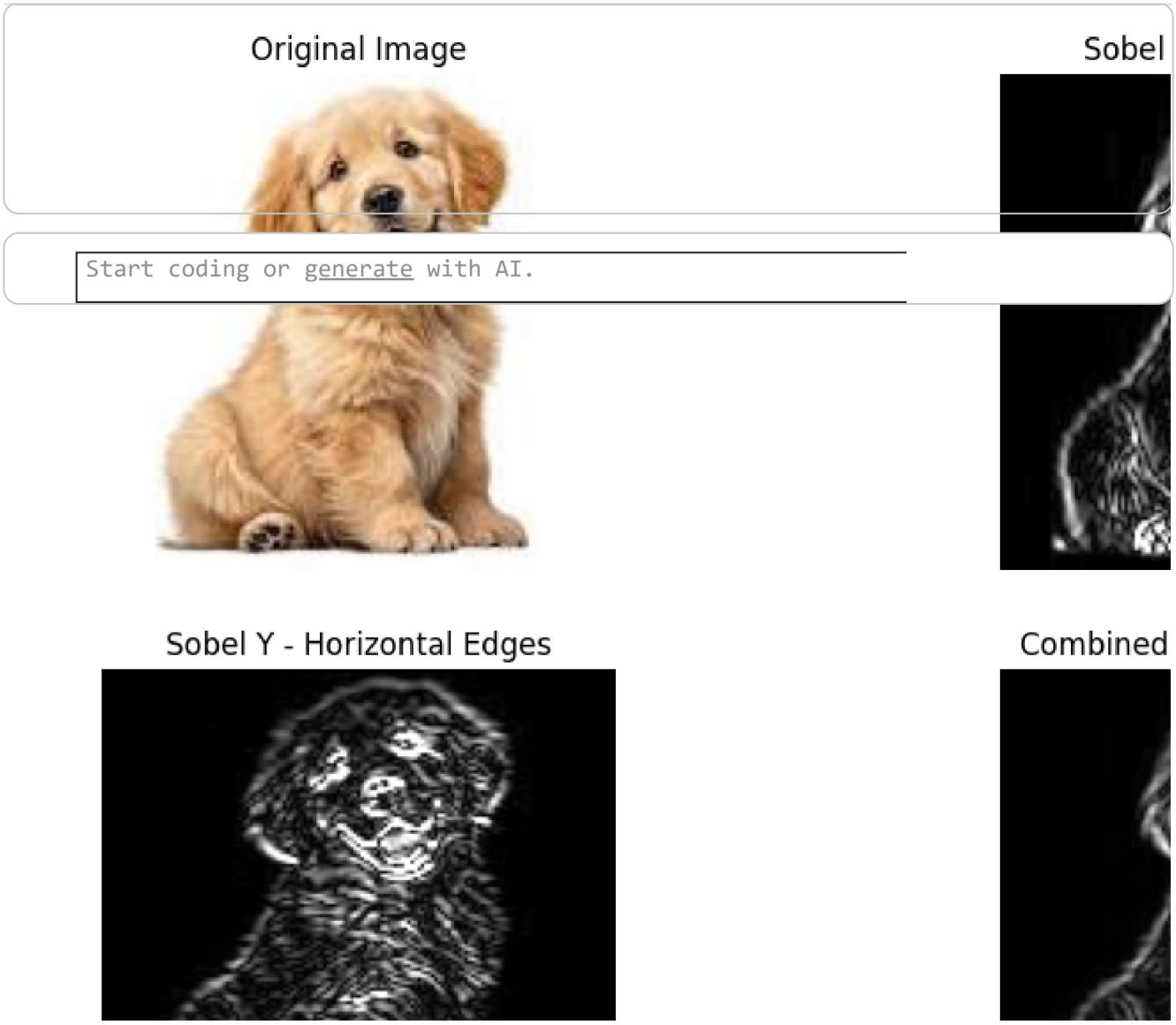

sobel_y = cv2.Sobel(
 gray,
 cv2.CV_64F,
 0,
 1,
 ksize=3
)
```

```
sobel_x = cv2.convertScaleAbs(sobel_x)
sobel_y = cv2.convertScaleAbs(sobel_y)

sobel_combined = cv2.addWeighted(
 sobel_x, 0.5,
 sobel_y, 0.5,
 0
)
```

Start coding or generate with AI.

```
plt.figure(figsize=(12, 8))
plt.subplot(2, 2, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis("off")
plt.subplot(2, 2, 2)
plt.imshow(sobel_x, cmap="gray")
plt.title("Sobel X - Vertical Edges")
plt.axis("off")
plt.subplot(2, 2, 3)
plt.imshow(sobel_y, cmap="gray")
plt.title("Sobel Y - Horizontal Edges")
plt.axis("off")
plt.subplot(2, 2, 4)
plt.imshow(sobel_combined, cmap="gray")
plt.title("Combined Sobel Edge Detection")
plt.axis("off")
plt.show()
```


Original Image
Sobel
Start coding or generate with AI.
Sobel Y - Horizontal Edges
Combined